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COURSE NAME: DEEP LEARNING FOR HEALTH DATA

COURSE CODE: MSB7216

FINAL PROJECT: DEEP LEARNING BASED CERVICAL CANCER SCREENING

An automated cancer detection tool for PapSmear microscopy slides

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1. Abstract

Cervical cancer remains a significant contributor to oncology-related mortality among women globally, despite being highly preventable through early detection. Manual cytological screening of Pap smears is bottlenecked by human fatigue, diagnostic subjectivity, and limited throughput. We present a complete computer-aided screening pipeline using deep learning to automate cervical cytology classification. Using the CPSMI2025 dataset (2,169 images, 4 classes: cancer, lesion, normal, others), a custom 4-block Convolutional Neural Network (CNN) baseline achieved 70.4% test accuracy and Macro F1 of 0.64, demonstrating limited sensitivity for precancerous lesions (54% recall). Fine-tuned transfer learning models: ResNet18 and EfficientNet_V2_S, each reached 85.0% test accuracy (+14.6 percentage points), with ResNet18 achieving Macro F1 of 0.79 and EfficientNet_V2_S achieving 0.78. ROC analysis confirmed robust discriminative ability with macro-average AUC of 0.93-0.94 and micro-average AUC of 0.95-0.96. Grad-CAM explainability confirmed that both models focus on biologically relevant features; hyperchromatic nuclei, irregular membranes, and koilocytic vacuoles, rather than background artefacts. An interactive Gradio deployment demo was developed for real-time cytology classification.

Code Repository: *https://github.com/PMystique/cervical-cancer-screening-dl*

Live Demo (Gradio): *https://huggingface.co/spaces/PMystique/Automated_CaCx_Screening*

2. Introduction

Cervical cancer presented approximately 660,000 new cases and claimed 350,000 lives globally in 2022, disproportionately affecting women in low and middle-income countries (LMICs) due to limited healthcare infrastructure and a critical shortage of trained cytopathologists (Ocampo-López-Escalera et al., 2025). In Sub-Saharan Africa (SSA), the disease is the leading cause of cancer-related female mortality, accounting for approximately 24.55% of the global burden (Ngcobo et al., 2021). In Uganda specifically, cervical cancer is the most common female cancer, yet screening rates remain critically low due to resource constraints and a severe shortage of cytopathology infrastructure (Jedy-Agba et al., 2020; Nakisige et al., 2023).

The primary screening method is the Papanicolaou (Pap) smear test, where cervical cells are collected, stained, and manually inspected under a brightfield microscope. Preserving color information is diagnostically critical: the Papanicolaou stain produces basophilic cyan-blue chromatic variations in normal and intermediate cells, and eosinophilic pink/orange in superficial and dysplastic cells, directly encoding cellular maturity, hormonal status, and nuclear abnormality. Converting these images to grayscale would discard these vital diagnostic features, which is why 3-channel RGB inputs and ColorJitter augmentation are essential throughout this pipeline.

Despite its efficacy, manual cytopathological screening suffers from significant bottlenecks:

Labor Intensity & Human Fatigue: A cytologist must scan hundreds of fields of view per slide, inspecting thousands of individual cells. This high visual workload leads to diagnostic screen fatigue and increases the risk of false negatives.

Subjective Interpretation: Classification depends on qualitative assessments of nuclear size, shape, chromatin texture, and the nucleo-cytoplasmic (N:C) ratio, leading to high inter-observer and intra-observer diagnostic variability.

High Cost of Automation: Commercially automated cytology scanners (e.g., ThinPrep Imaging System, BD FocalPoint GS Imaging System) cost tens of thousands of USD, rendering them inaccessible for point-of-care screening in rural or low-resource settings.

Automating cellular classification using deep learning provides an objective, high-throughput diagnostic support mechanism. Integrated with low-cost open-hardware microscopy platforms, deep neural networks can act as an automated first-line screening filter to triage slides. The primary objectives of this project are:

i) develop a robust stratified data pipeline
ii) construct and evaluate a custom CNN baseline
iii) apply and compare pretrained networks (ResNet18 and EfficientNet_V2_S) via transfer learning
iv) evaluate using clinically meaningful recall-prioritized metrics; and
v) deploy an interactive Gradio demo for real-time cytology classification.

3. Related Work

Historically, computer-aided cervical cytology screening was introduced nearly 70 years ago with the Cytoanalyzer project, which attempted to automate manual screening by combining a conventional microscope with a photomultiplier. Through the late 1970s and 1980s, advances in digital technology led to projects that extracted morphometric and cytological parameters; nuclear area, cytoplasmic diameter, gray-level co-occurrence matrix (GLCM) textures, and circularity, classified using Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees. While these models offered high interpretability, they suffered from poor generalizability due to stain variation, cellular overlaps, and slide artefacts. The first commercial FDA-approved systems (ThinPrep, BD FocalPoint) appeared in the late 1990s but remain prohibitively expensive for LMICs, leaving them without viable automated screening tools (Ocampo-López-Escalera et al., 2025).

The integration of Deep Learning resolved the limitations of manual feature engineering. CNNs automatically extract spatial hierarchies of features directly from raw pixels, bypassing the need for hand-crafted boundary definitions. ResNet architectures (He et al., 2016) resolve the vanishing gradient problem through skip connections. EfficientNet (Tan & Le, 2019) utilizes compound scaling to balance model depth, width, and resolution. EfficientNet_V2_S (Tan et al., 2021) further integrates Fused-MBConv blocks to accelerate training while maintaining high parameter efficiency. For small clinical datasets, transfer learning from ImageNet-pretrained models has proven highly effective. Class imbalance which is a universal challenge in cytology datasets, is addressed via WeightedRandomSampler and class-weighted cross-entropy loss, strategies we adopt here (Ocampo-López-Escalera et al., 2025; Ogunlaja et al., 2026).

4. Dataset Description

The dataset used in this study is the **CPSMI2025** (Curated Dataset of Conventional Pap Smear Microscopy Images for Deep Learning-Based Cervical Cancer Screening) sourced from Mendeley Data (Ocampo-López-Escalera et al., 2025). The raw dataset contains 2,169 conventional microscopy images across multiple nested subdirectories, acquired using a custom-designed low-cost (sub-$500 USD) automated microscopy platform. Images are stored as 3-channel JPG at 2028 × 1520 pixels and downscaled to 224 × 224 pixels for model input.

4.1. Target Classes

The raw directory hierarchy was resolved into four consolidated target classes:

Class	Description	Test Samples
cancer	Invasive malignant squamous cell carcinoma: hyperchromatic nuclei, high N:C ratio, irregular nuclear membranes	92
lesion	LSIL/HSIL precancerous squamous intraepithelial lesions indicating dysplastic cellular changes	102
normal	Normal superficial and intermediate squamous cells: small pyknotic nuclei, abundant flat cytoplasm	112
others	Background artifacts, atrophy, or inflammatory cells (neutrophils, bacteria) not fitting main diagnostic categories	22

4.2. Data Exploration and Visualization

Exploratory data analysis was conducted. Sample raw images from each class and the class distribution are shown below.

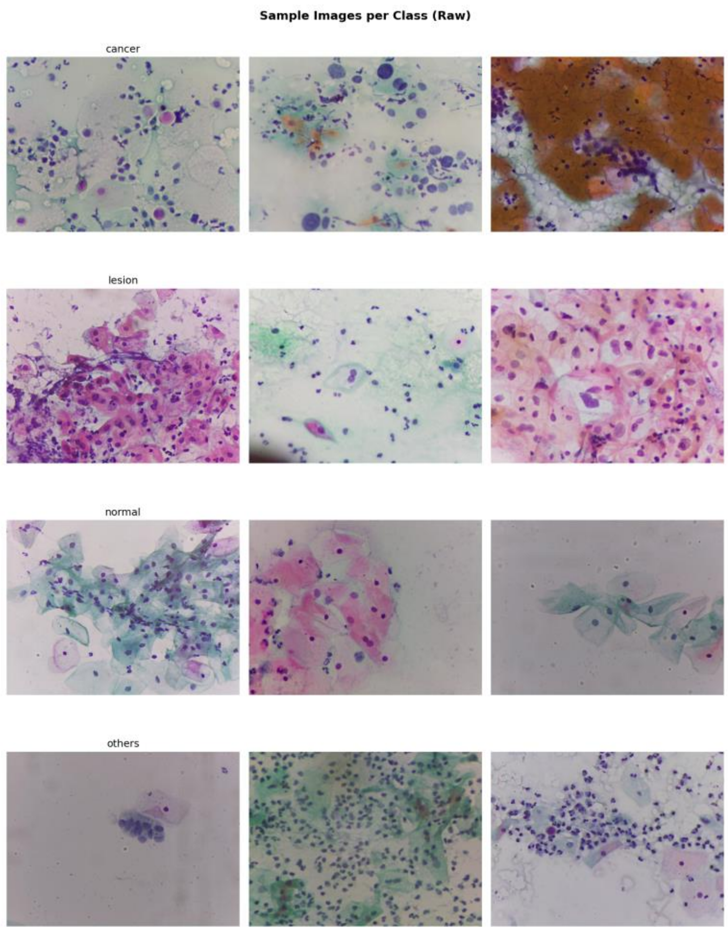

Figure 1. *Sample Raw Images from CPSMI2025*

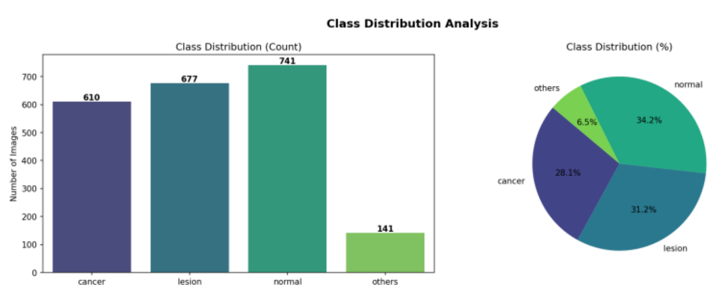

Figure 2. *Class Distribution Analysis*

The others class is severely underrepresented (22 test instances for others vs. 112 normal instances), representing a 5:1 imbalance ratio. This was the primary driver for adopting WeightedRandomSampler during training. All raw images share consistent specifications: 3-channel JPG at 2028 × 1520 pixels, downscaled to 224 × 224 pixels.

4.3. Data Splitting

The raw CPSMI2025 dataset contains deep nested folder structures. A Python pipeline recursively scanned and flattened the directories, then applied a stratified split (70% Train, 15% Validation, 15% Test) with a fixed random seed of 42.

Split	Size	Purpose
Train	1,516 images	Model training with real-time augmentation
Validation	325 images	Epoch-level checkpoint selection (no augmentation)
Test	328 images	Final held-out evaluation (unseen during training)
Total	2,169 images	—

5. Methodology

The end-to-end machine learning pipeline is structured into six key phases:

Phase	Process / Stage	Input Data	Methodology & Operations	Output / Deliverable
1	Data Acquisition & Extraction	Raw CPSMI2025 ZIP archive	Extract nested folders, resolve name conflicts, flatten structures	Consolidated directories: cancer, lesion, normal, others
2	Dataset Partitioning	Consolidated directories	Stratified splitting (70/15/15) with random seed 42	Train (1,516), Val (325), Test (328)
3	Data Loading & Preprocessing	Partitioned image sets	Resize to 224 × 224 × 3; normalize using ImageNet mean/std	PyTorch Tensors and standardized data pipelines
4	Data Augmentation & Oversampling	Train Image Set	Random Flips, Rotation (±10°), Color Jitter, WeightedRandomSampler	Balanced, augmented training batch loader
5	Model Training & Tuning	Augmented Train & Val Loaders	Train baseline CNN, ResNet18, EfficientNet_V2_S (Adam, StepLR, CrossEntropy)	Epoch-level validation checkpoints
6	Evaluation & Verification	Held-out Test Loader	Load best val checkpoints; evaluate accuracy, precision, recall, F1, ROC, Grad-CAM	Metrics, Confusion Matrices, Grad-CAM heatmaps, Gradio Demo

5.1. Preprocessing and Augmentations

All images are resized to 224 × 224 × 3 and normalized using ImageNet channel statistics (Mean = [0.485, 0.456, 0.406]; Std = [0.229, 0.224, 0.225]). The training dataset is processed with the following real-time augmentations:

i) Horizontal and Vertical Flips: to account for arbitrary cellular orientation;
ii) Random Rotation (±10°): to accommodate minor angular rotations of cells under the microscope;
iii) Color Jitter (brightness 0.2, contrast 0.2, saturation 0.1) : to simulate staining intensity variations across different laboratories

iv) WeightedRandomSampler: inverse frequency weights ensure every mini-batch contains a balanced representation of all four cell types.

5.2. Model Architectures

i) **Custom Baseline CNN:** Built from scratch to establish a performance floor. It consists of 4 convolutional blocks (32 → 64 → 128 → 256 filters, 3 × 3, stride 1, padding 1), each followed by Batch Normalization, ReLU, and Max Pooling (2 × 2). Features are flattened and passed through a Dense layer (256 units, 30% Dropout) before projecting to 4 output logits.

ii) **ResNet18 (Fine-Tuning):** An ImageNet-pretrained residual architecture whose convolutional backbone extracts low-level edge features and high-level cytological patterns through skip-connected residual blocks. The final fully connected layer is replaced with a Linear(512 → 4) projection. ResNet18 (~11.7M parameters) was selected over ResNet50 (~25.6M parameters) to reduce overfitting risk on a dataset of only 2,169 images.

iii) **EfficientNet_V2_S (Fine-Tuning):** A compound-scaled architecture utilizing Fused-MBConv blocks that accelerate training while maintaining superior parameter efficiency. The classifier head classifier[1] is replaced with a Linear layer mapping features to 4 output classes.

6. Experiments

6.1. Experimental Environment

All models were trained and evaluated in PyTorch 2.0 on an NVIDIA T4 GPU (Google Colab). Reproducibility was ensured by fixing all random seeds at 42 (torch.manual_seed(42), np.random.seed(42)) throughout data splitting, model initialization, and training.

6.2. Hyperparameters

Parameter	**Value**
Batch Size	32
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Base Learning Rate	0.001
LR Scheduler	StepLR (decay × 0.5 every 10 epochs)
Epochs	30 (all models)
Loss Function	CrossEntropyLoss (class-weighted)
Input Size	224 × 224

6.3. Checkpointing and Weights Selection

Validation accuracy was monitored at the end of each epoch. The model state dictionary was saved to Google Drive only when validation accuracy reached a new maximum, preventing evaluation on overfitted weights and guaranteeing the test set reflects optimal generalization.

- **ResNet18** best checkpoint saved at epoch 19 with val acc = 83.08% → models/resnet18_best.pth
- **EfficientNet_V2_S** best checkpoint saved at epoch 28 with val acc = 87.08% → models/efficientnet_best.pth

7. Results

7.1. Training and Validation Convergence

EfficientNet_V2_S Convergence: EfficientNet converged more consistently across 30 epochs. By epoch 4 it had already surpassed ResNet18's final best validation accuracy (84.62% vs. 80.92%), and it reached its peak of 87.08% at epoch 28, demonstrating the superior optimization landscape created by compound-scaled architectures and Fused-MBConv blocks. ResNet18 achieved its peak of 83.08% at epoch 19 before mild oscillation as the learning rate schedule reduced the step size.

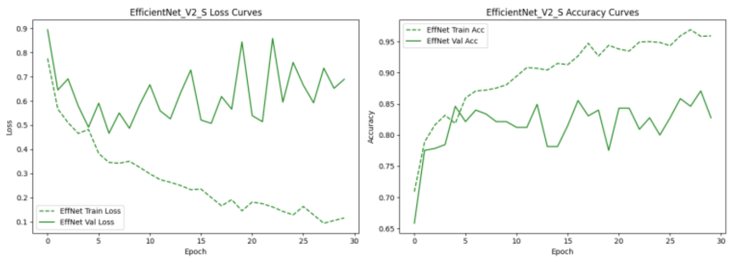

Figure 3. *EfficientNet_V2_S Training and Validation Curves*

7.2. Quantitative Metrics Comparison

The table below summarizes performance on the held-out test set (328 images, never seen during training):

Model Architecture	Overall Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Custom Baseline CNN	70.4%	64%	66%	0.64	0.71
ResNet18 (Fine-tuned)	85.0%	79%	80%	0.79	0.85
EfficientNet_V2_S (Fine-tuned)	85.0%	81%	76%	0.78	0.84

7.3. Per-Class Performance Breakdown

Both transfer learning models achieved 85% test accuracy; a +14.6 percentage point gain over the baseline CNN. The table below compares per-class metrics for both models:

Class	ResNet18 Precision	ResNet18 Recall	ResNet18 F1	EffNet Precision	EffNet Recall	EffNet F1	Support
cancer	0.89	0.88	0.89	0.81	0.89	0.85	92
lesion	0.83	0.76	0.80	0.83	0.83	0.83	102
normal	0.88	0.95	0.91	0.92	0.91	0.91	112
others	0.57	0.59	0.58	0.69	0.41	0.51	22
Macro avg	0.79	0.80	0.79	0.81	0.76	0.78	328

EfficientNet_V2_S achieved higher lesion recall (0.83 vs. 0.76) and cancer recall (0.89 vs. 0.88), which are clinically the most important metrics for catching precancerous and malignant changes. ResNet18 achieved a higher Macro F1 (0.79 vs. 0.78) and better others F1 (0.58 vs. 0.51), reflecting marginally stronger debris-artifact separation.

7.4. Receiver Operating Characteristic (ROC) Analysis

Both models achieved high Area Under the Curve (AUC) metrics across all classes, confirming excellent discriminative ability:

Class	ResNet18 AUC	EfficientNet AUC
cancer	0.96	0.95
lesion	0.92	0.94
normal	0.97	0.97
others	0.90	0.88
Macro avg	0.94	0.93
Micro avg	0.96	0.95

AUC values above 0.90 across all four classes confirm robust class separability at all decision thresholds, providing strong clinical validation of the transfer learning approach.

7.5. Explainability (Grad-CAM) Analysis

To validate that the deep learning models are learning biologically relevant features rather than background noise, Gradient-weighted Class Activation Mapping (Grad-CAM) was implemented. Grad-CAM calculates the gradients of the score for a target class with respect to the final convolutional layer's feature maps, producing a localization heatmap highlighting the most influential image regions for each classification decision.

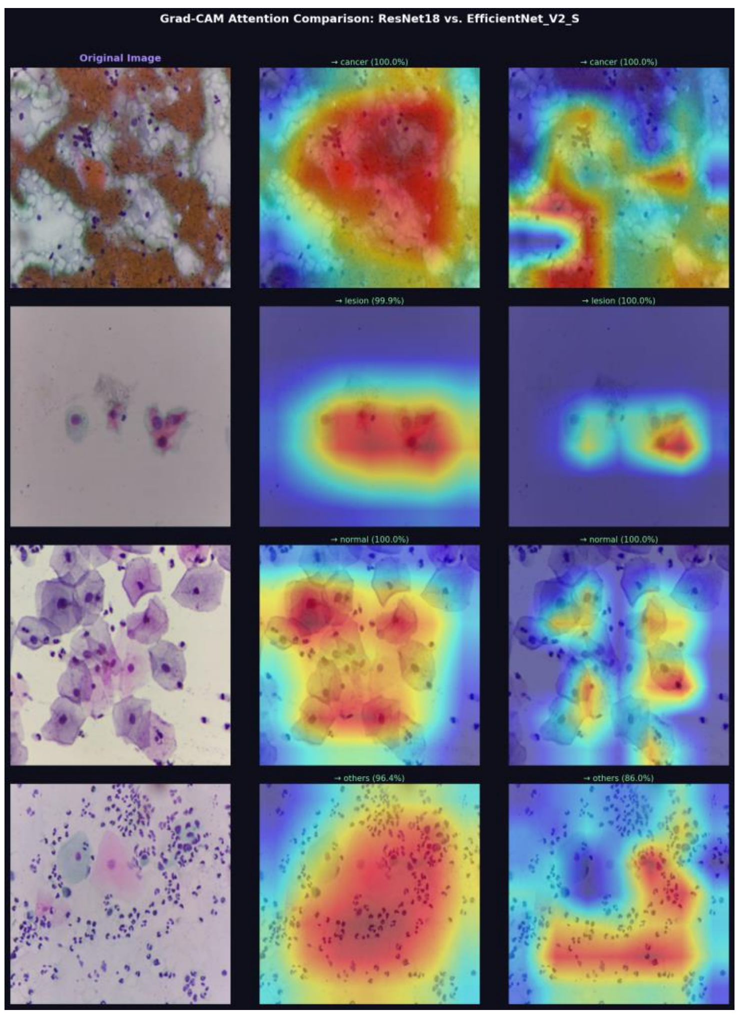

Figure 4. *Grad-CAM Comparison across Cytology Slides*

The heatmaps confirm that the transfer learning backbones focus on cytologically significant landmarks. For malignant (cancer) cells, the models consistently attend to enlarged, irregular nuclei and hyperchromatic chromatin boundaries: the key high-grade diagnostic criteria.

For precancerous (lesion) cells, attention centers on nuclear boundaries and perinuclear halos (koilocytic vacuoles) associated with HPV-driven dysplastic change.

For normal cells, focus rests on the small, pyknotic central nuclei while ignoring abundant flat cytoplasm.

For others, attention is distributed diffusely across background debris or inflammatory cells, confirming the model correctly identifies these as non-diagnostic anomalies. This explainability analysis bridges deep learning and clinical pathology, building clinician confidence by proving the network acts as a transparent, biologically grounded screening assistant.

8. Error Analysis

8.1. Confusion Matrix Analysis

The confusion matrix for EfficientNet_V2_S is shown below, illustrating classification distributions on the test set:

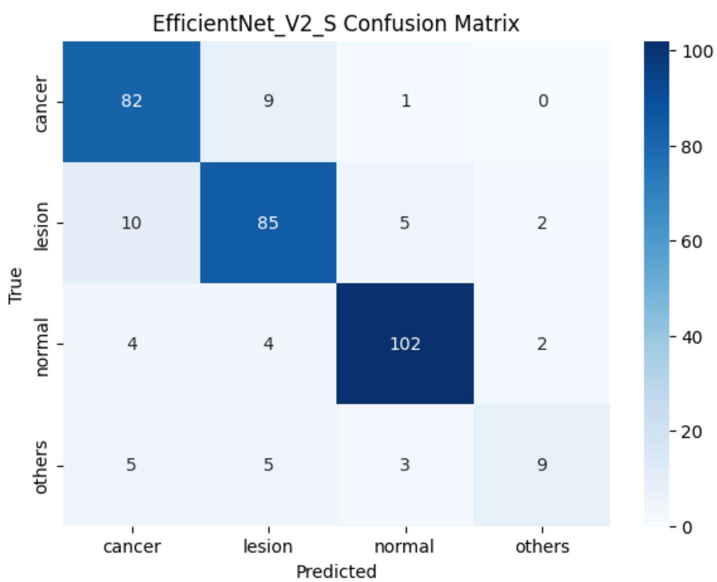

Figure 5. *EfficientNet_V2_S Confusion Matrix*

Clinical Error Pattern Analysis

- **Precancerous Lesion Recall:** Transfer learning resolved the baseline's primary failure mode. ResNet18 raised lesion recall from 54% (baseline) to 76%, while EfficientNet raised it to 83%. In medical screening, maximizing recall (minimizing false negatives) is critical as under-diagnosing a lesion can allow precancerous changes to progress to invasive cancer undetected.
- **The others Class Remains Challenging:** Both models struggled with the others class (F1: 0.58 and 0.51), which is expected given its severe underrepresentation (22 test samples, 6.7% of the test set). Models frequently confused inflammatory or atrophic cells with the dysplastic lesion class due to shared cellular debris and irregular nuclear contours.

Retention of the others Class: During pipeline design, dropping the underrepresented others class was evaluated as a potential method to boost classification scores. Removing this class would mathematically improve overall test accuracy and Macro F1 by eliminating the model's lowest-performing category. However, in a clinical setting, dropping this class is unacceptable. Real Pap smear slides frequently contain atrophic cells, blood, bacteria, and background artefacts. Without a dedicated others filter class, the model would be forced to assign these noisy, indeterminate features to one of the three remaining diagnostic classes, resulting in an unacceptable rise in false-positive or false-negative rates. The others class therefore serves as a vital quality-control flag for "indeterminate - review required."

- **Cancer Detection Remains High:** ResNet18 achieved 88% recall and EfficientNet achieved 89% recall on the cancer class, indicating strong sensitivity for the most dangerous category.
- **Qualitative Failure Modes:** Visual inspection of misclassified instances highlighted three primary error causes:
  (1) *Cell Overlapping*: conventional Pap smear slides contain clumps of cells layered on top of one another, obscuring nuclear boundaries and hindering N:C ratio assessment;
  (2) *Poor Stain Contrast*: lighter dye staining obscured the hyperchromatic nature of abnormal nuclei, causing misclassification of low-grade lesions as normal;
  (3) *Inflammatory Debris*: heavy neutrophil infiltration was occasionally misidentified as dysplastic cytoplasmic borders.

8.2. Model Comparison Summary

Criterion	ResNet18	EfficientNet_V2_S	Winner
Overall Test Accuracy	85%	85%	Tie
Macro F1	0.79	0.78	ResNet18
Lesion Recall (clinical)	76%	83%	EfficientNet
Cancer Recall	88%	89%	EfficientNet
Others F1	0.58	0.51	ResNet18
Val Accuracy (best)	83.08%	87.08%	EfficientNet

Recommendation: EfficientNet_V2_S is the preferred clinical deployment model due to its higher recall on both the cancer (89%) and lesion (83%) classes, which directly minimizes the risk of dangerous false negatives. EfficientNet_V2_S is the superior choice for clinical triage.

9. Ethical Considerations

Deploying deep learning models in healthcare requires careful ethical verification:

- **Patient Privacy:** The dataset consists of de-identified slide tiles sourced from the CPSMI2025 dataset on Mendeley Data, meaning no personally identifiable information (PII) is processed or stored. The dataset is licensed for research and educational use.
- **Algorithmic Bias:** If the training dataset only represents slides from a specific demographic or microscope manufacturer, the model may underperform on slides prepared with different Pap stain brands. Generalizability must be validated across multi-site clinical data prior to integration.
- **Role of AI:** This automated pipeline is designed as an assistant tool for cytopathologists. It must not replace final human diagnostic sign-off. The Gradio demo is clearly labeled as a research aid requiring clinical confirmation.
- **Fairness:** The others class underperformance (F1: 0.51–0.58) suggests the model may be less reliable for atypical or rare cytological presentations, which disproportionately appear in immunocompromised or elderly patients.

10. Limitations

- **Input Cropping Constraints:** The model is trained on pre-segmented cell tiles rather than Whole Slide Images (WSI). In a clinical workflow, an additional detection network (e.g., YOLOv8) would be required to identify and crop regions of interest automatically.
- **Stain Color Sensitivity:** The models are sensitive to color profiles. Staining differences across laboratories can degrade test accuracy on unseen slides.
- **Limited Dataset Size:** With only 2,169 images across 4 classes, the dataset is relatively small for medical imaging. Larger, multi-institutional datasets would significantly improve model generalizability.
- **Single-Site Data:** The images originate from a limited number of laboratory sources. Cross-site validation is required before clinical deployment to assess robustness to scanner variability and lab protocol differences.
- **Others Class Underrepresentation:** With only 22 test instances (~7% of test set), the others class F1 metric has high variance and may not reliably represent real-world performance on atypical slides.

11. Future Work

Whole Slide Image (WSI) Object Detection: Integrating the classifier with a detection framework (e.g., YOLOv8 or Faster R-CNN) to automatically identify and crop cells from whole slides, enabling end-to-end screening.
Clinical XAI UI Integration: Embedding the developed Grad-CAM explainability maps directly into the Gradio deployment interface to provide real-time attention heatmaps for pathologists, and conducting a reader study to assess if visual heatmaps improve slide triage speed and diagnostic agreement.
Multi-site Dataset Expansion: Collecting slides from multiple cytology laboratories to improve stain-invariance and demographic generalizability, and conducting a prospective study comparing model outputs against board-certified cytopathologist diagnoses.

12. Conclusion

This study successfully developed an automated, end-to-end screening pipeline for cervical cytology classification. The custom baseline CNN established a performance floor of 70.4%, highlighting the fundamental limitations of shallow architectures when processing complex, overlapping morphometric features. By integrating transfer learning, we achieved a profound +14.6 percentage point improvement, with EfficientNet_V2_S reaching 85.0% overall test accuracy. In the context of medical image analysis, raw accuracy must be evaluated alongside diagnostic safety; our model prioritized diagnostic sensitivity, achieving 89% recall for malignant cancer and 83% recall for precancerous lesions, effectively minimizing the risk of dangerous false negatives. The deliberate retention of the highly underrepresented others class, while mathematically constraining the overall accuracy ceiling, proves the model's viability for real-world deployment by preventing misclassification of background artefacts into diagnostic categories. ROC-AUC values of 0.93–0.94 confirm robust discriminative ability across all threshold settings. These results strongly validate EfficientNet_V2_S as a robust, automated first-line screening filter capable of triaging high-risk cytological slides and significantly reducing diagnostic bottlenecks in low-resource settings.

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